Vishnu Waman Thakur Charitable Trust's

VIVA Institute of Technology

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Affiliated to University of Mumbai

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Semester _____Academic Year _____has completed Term work (Practicals / Tutorials / Assignments /Test)in Subject_____ as per the syllabus prescribed by the University of Mumbai successfully.

Faculty

Mini Project on

" FinSent – Financial Sentiment Analysis for Market Trend Prediction "

Submitted in partial fulfillment of the requirements of the degree

of

Bachelor of Engineering in Computer Science & Engineering (AI & ML) Department

Submitted by

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

(AI & ML) VIVA INSTITUTE OF TECHNOLOGY

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2026-27

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INTRODUCTION

The stock market is fundamentally driven by two factors: mathematical data (prices, volumes, financials) and human emotion (news, rumors, and sentiment). In the modern digital era, thousands of financial news articles are published daily. Parsing this massive volume of unstructured text to determine whether the market sentiment for a specific asset is bullish (positive) or bearish (negative) is impossible for a human trader to do in real-time.

FinSent is an advanced Natural Language Processing (NLP) and Financial Technology (FinTech) web application designed to solve this problem. It autonomously fetches real-time market data and news, processes the textual data using Large Language Models (LLMs), and outputs a highly structured dashboard. The application does not just classify news as positive or negative; it uses generative AI to extract key actionable insights, perform automated SWOT (Strengths, Weaknesses, Opportunities, Threats) analyses, and generate peer comparisons, bridging the gap between raw data and actionable trading intelligence.

MOTIVATION

Institutional investors (hedge funds, investment banks) spend tens of thousands of dollars annually on proprietary software like Bloomberg Terminals, which possess built-in AI tools to analyze news sentiment instantly. Retail (individual) investors lack access to these expensive resources, placing them at a severe disadvantage due to "Information Overload."

Standard, traditional NLP algorithms (like VADER or TextBlob) are fundamentally inadequate for financial markets. In finance, words have unique contexts (e.g., "The Federal Reserve *cut* rates" is generally highly positive for stocks, whereas "cut" is usually negative in standard English). This motivated the creation of a specialized pipeline that leverages modern, context-aware Large Language Models to accurately interpret complex financial jargon and empower retail investors with institutional-grade insights for free.

OBJECTIVES

The primary objectives of the FinSent project are:

1) **Automated NLP Sentiment Analysis:** To accurately classify real-time financial news articles regarding specific stock tickers into Positive, Negative, or Neutral sentiments.
2) **Actionable Information Extraction:** To prompt an LLM to read full articles and extract the single most important "highlight" sentence, saving the user's reading time.
3) **Generative Fundamental Analysis:** To automatically generate a comprehensive AI SWOT Analysis and Peer Comparison table based on the company's real-time financial metrics.
4) **Advanced UI/UX:** To display this data on a highly responsive, aesthetically advanced Glassmorphism dashboard featuring dynamic theming (Light/Dark mode) and contextual color-coding.

RESOURCES & TECHNOLOGY STACK USED

The project utilizes a modern, lightweight, and highly efficient technology stack, enforcing a strict separation of concerns between the backend processing and frontend rendering.

- **Backend Framework:** Python 3.x with **FastAPI** (for high-performance, asynchronous RESTful API endpoints).
- **Data Aggregation:** yfinance library (to fetch live ticker prices, historical data, fundamental ratios, and Yahoo Finance news headlines).
- **NLP & LLM Engine: OpenRouter API** utilizing the google/gemma-4-26b-a4b-it:free model for text processing, sentiment classification, and generation of structured JSON responses.
- **Frontend Structure & Logic:** Vanilla HTML5 and Vanilla JavaScript (ES6+).
- **Frontend Styling:** Pure CSS3 utilizing advanced features such as CSS Variables (for Light/Dark theming), Flexbox/Grid layouts, @keyframes micro-animations, and Glassmorphism effects (backdrop-filter).
- **Data Visualization: Chart.js** (for rendering dual-axis hybrid Line/Bar charts with Moving Average overlays).
- **Iconography:** FontAwesome CDN.

ALGORITHM & PROCESSING LOGIC

The core logic of the application operates on a strict request-response lifecycle driven by asynchronous NLP processing:

i. **Input Acquisition:** The user enters a stock ticker (e.g., AAPL) and selects the relevant global market exchange via the frontend UI.

ii. **Data Fetching:** The backend receives the API request and utilizes yfinance to scrape the latest news articles and historical price data for the requested ticker.

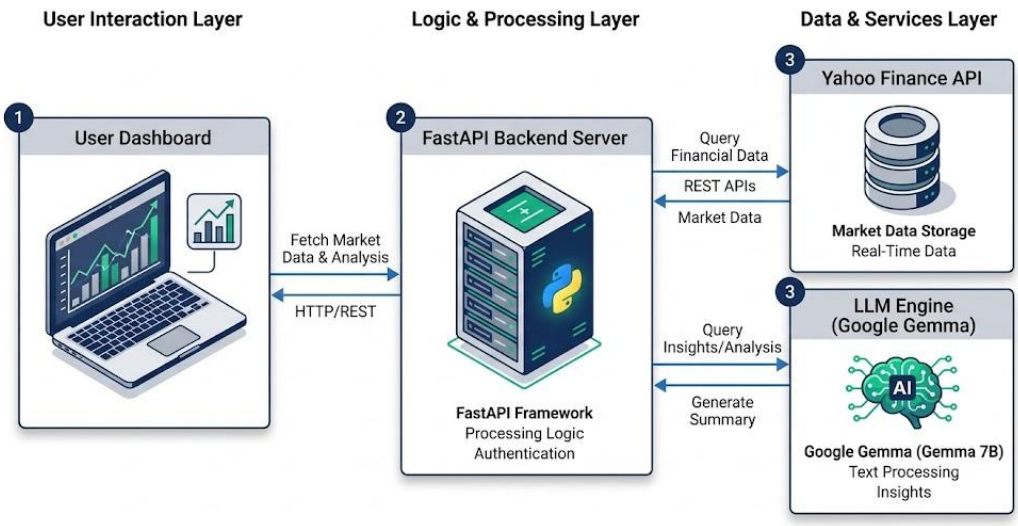
iii. **Prompt Engineering (NLP Phase):** The backend constructs a rigorous prompt containing the news headlines and fundamental data. It instructs the Gemma LLM to:

- Evaluate the overall sentiment.
- Perform Entity Recognition to suggest competitor peers.
- Formulate a SWOT matrix.
- Extract exactly one sentence as a key "highlight" per news article.

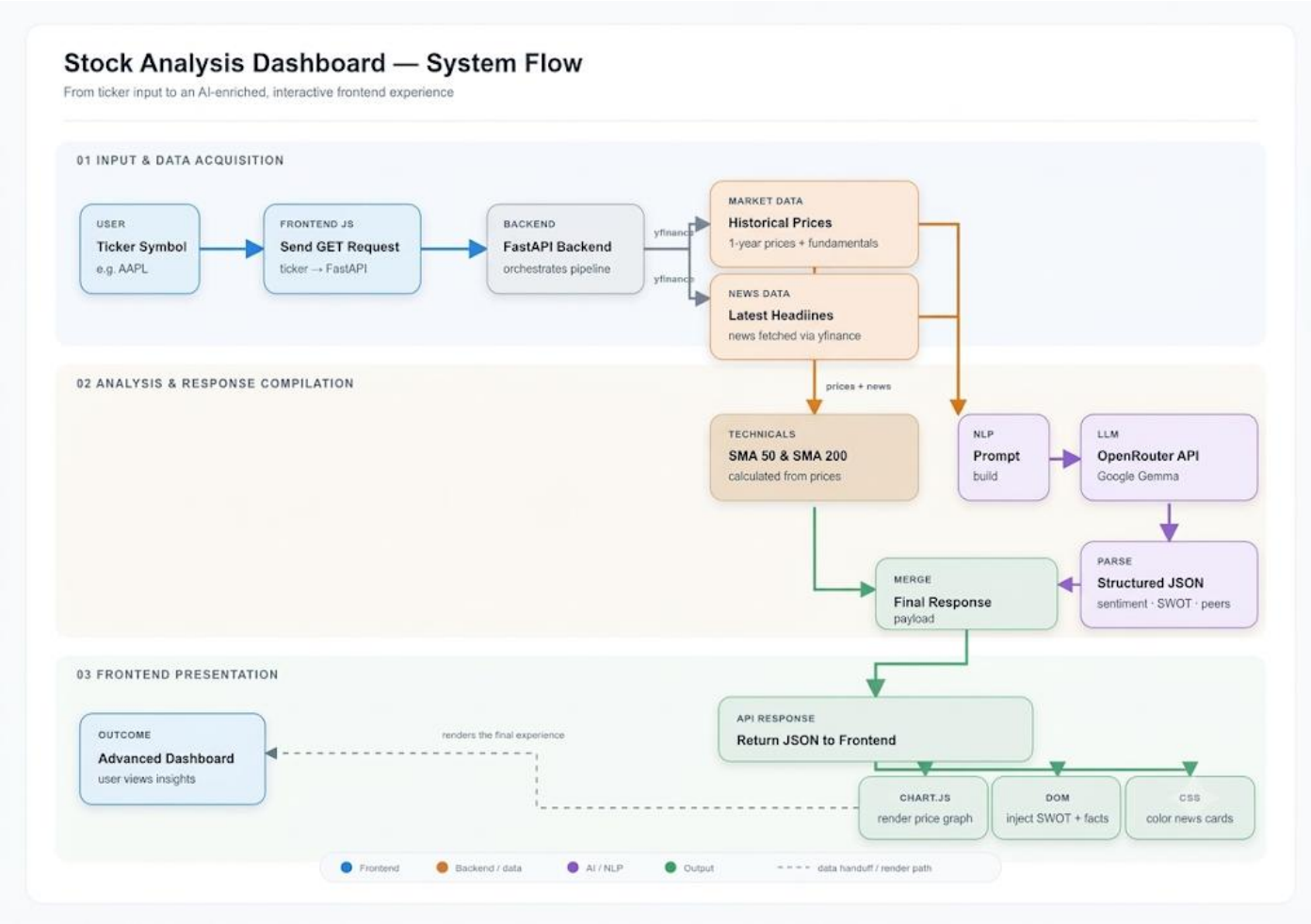
iv. **JSON Enforcement:** The LLM is strictly prompted to return the output in a parseable JSON format.

v. **Data Hydration:** The backend parses the LLM's JSON response, merges it with the calculated technical indicators (SMA 50, SMA 200), and returns the payload to the frontend.

vi. **Dynamic Rendering:** The vanilla JavaScript engine receives the payload, dynamically injects the data into the HTML DOM, updates the Chart.js instance, and applies contextual CSS classes (e.g., coloring a news highlight box red if the sentiment is NEGATIVE).



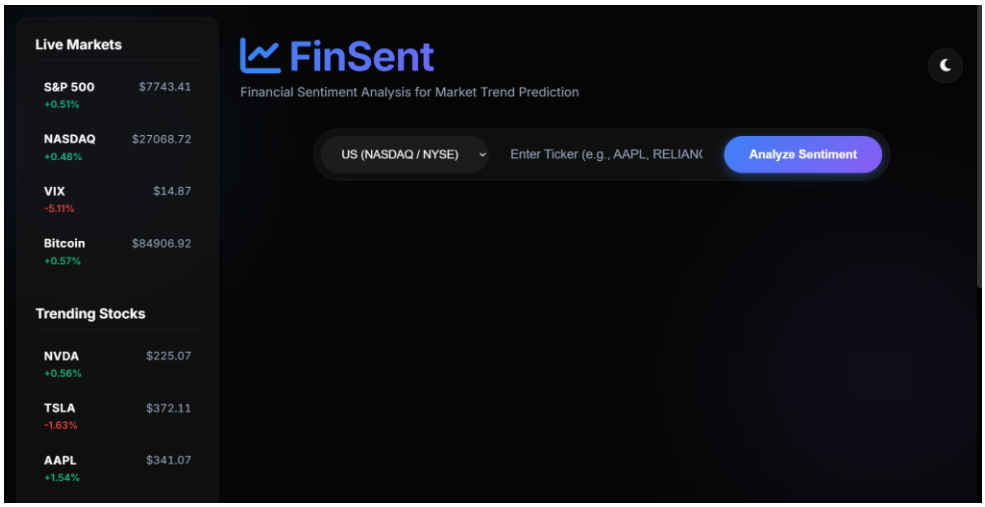
FLOWCHART

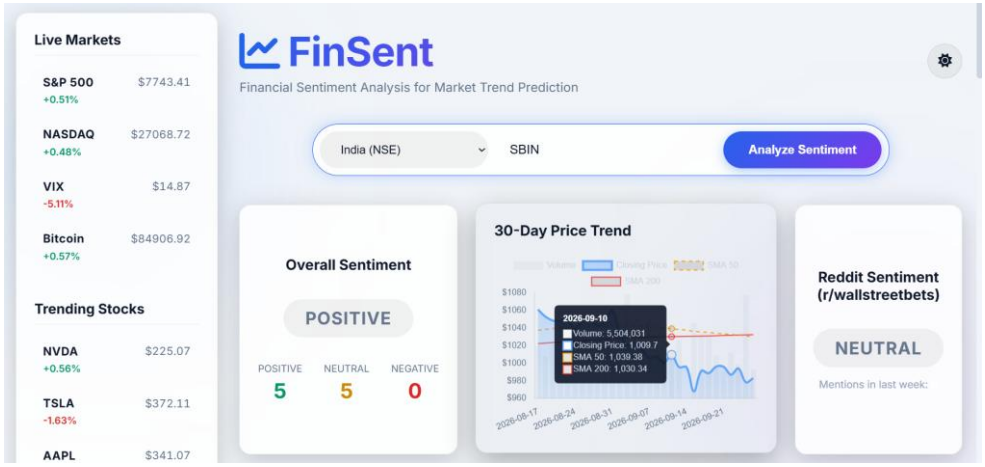

Stock Analysis Dashboard — System Flow
From ticker input to an AI-enriched, interactive frontend experience
01 INPUT & DATA ACQUISITION
USER
Ticker Symbol
e.g. AAPL
FRONTEND JS
Send GET Request
ticker → FastAPI
BACKEND
FastAPI Backend
orchestrates pipeline
yfinance
yfinance
MARKET DATA
Historical Prices
1-year prices + fundamentals
NEWS DATA
Latest Headlines
news fetched via yfinance
02 ANALYSIS & RESPONSE COMPILATION
prices + news
TECHNICALS
SMA 50 & SMA 200
calculated from prices
NLP
Prompt
build
LLM
OpenRouter API
Google Gemma
PARSE
Structured JSON
sentiment · SWOT · peers
MERGE
Final Response
payload
03 FRONTEND PRESENTATION
API RESPONSE
Return JSON to Frontend
CHART.JS
render price graph
DOM
inject SWOT + facts
CSS
color news cards
renders the final experience
OUTCOME
Advanced Dashboard
user views insights
Frontend
Backend / data
AI / NLP
Output
data handoff / render path

OUTPUT & INTERFACE BRIEFING

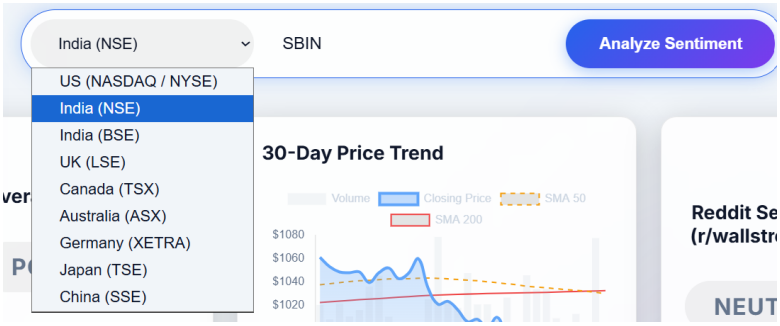
The user interface is engineered to mimic high-end institutional trading software. Key interface components include:

✓ **The Market Sidebar:** A real-time, persistent widget displaying overall market health (S&P 500, NASDAQ, Bitcoin), trending retail stocks, and sector ETF performances.



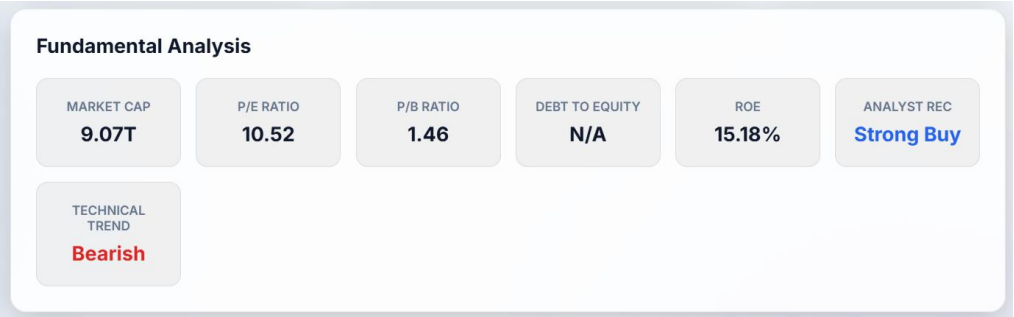


✓ **Search & Theme Header:** A centralized search bar with an integrated dropdown for global exchanges. A theme toggle switch dynamically switches the entire CSS variable scope between a "Pitch Black" neon dark mode and a sleek light mode.



✓

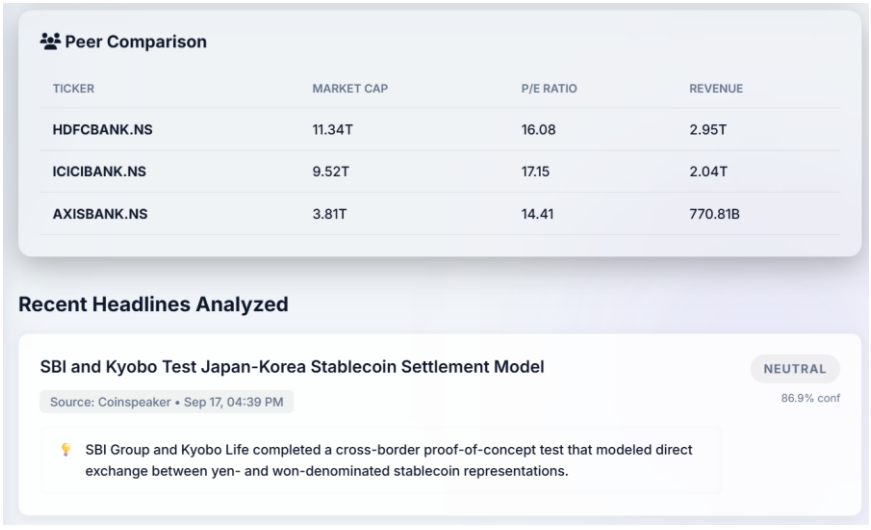
✓ **Summary & Charting:** Displays the aggregate sentiment badge alongside a 30-day interactive price/volume chart overlaid with SMA 50 and SMA 200 trendlines.

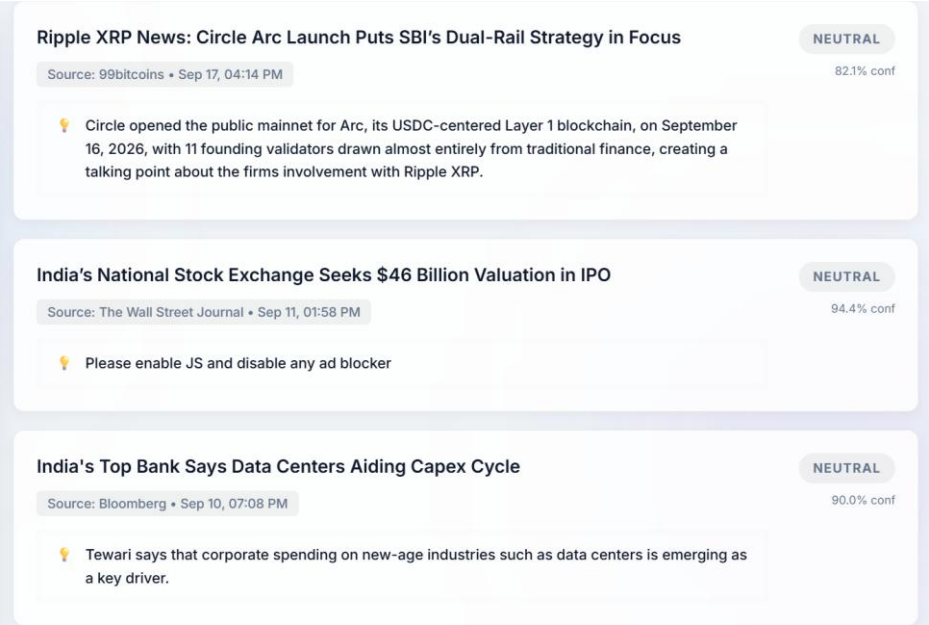


✓ **V2 Analytical Dashboard:**

- *Fundamentals Grid:* Displays Market Cap, P/E, Debt to Equity, and Analyst Recommendations.
- *AI SWOT Analysis:* A 4-quadrant colored grid generated by the LLM detailing the company's internal strengths and external threats.
- *Peer Comparison Table:* Automatically identifies competitors and compares their market cap and revenue.







✓ **Contextual News Feed:** Displays recent headlines with AI-extracted "hint" boxes. The highlight boxes dynamically change their background and border colors (Green, Red, Yellow) based on the LLM's classification of the article's sentiment.



CONCLUSION & FUTURE SCOPE

Conclusion

The FinSent project successfully demonstrates how Natural Language Processing and generative LLMs can be integrated into a lightweight web application to solve real-world financial problems. By automating the extraction of sentiment, SWOT analyses, and key highlights, the application effectively mitigates information overload for retail investors. The project achieved a robust, responsive, and aesthetically advanced dashboard without relying on heavy frontend frameworks, relying entirely on optimized Vanilla JS and CSS.

Future Scope

While highly functional, the system has several avenues for future expansion:

1. **Alternative Institutional Data (SEC Filings):** Expanding the NLP pipeline to automatically scrape and analyze the sentiment of SEC 10-K/10-Q EDGAR filings and Quarterly Earnings Call audio transcripts.
2. **Real-Time WebSockets:** Replacing standard REST API calls with WebSockets to stream live stock prices and breaking news alerts instantly to the dashboard without requiring a page refresh.
3. **Automated Trading Webhooks:** Connecting the sentiment analysis pipeline to a broker API (like Alpaca or Zerodha) to autonomously execute paper trades based on sudden spikes in highly positive or highly negative news sentiment.